

Distributed Microservices Bases E-Commerce Platform

Database Schema

There are total 3 tables

1. Users

Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each user
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password	VARCHAR(255)	NOT NULL	Encrypted password
email	VARCHAR(100)	UNIQUE, NOT NULL	User email
role	ENUM	('customer','vendor','admin') NOT NULL	Defines user type
phone	VARCHAR(10)	UNIQUE ,NOT NULL	Customer/vendor phone number
address	VARCHAR(255)	NOT NULL	Customer shipping address / Vendor business address
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

2. Products

Column Name	Data Type	Data Type	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each product
name	VARCHAR(100)	NOT NULL	Product name
description	TEXT	NOT NULL	Product description
price	DECIMAL(10,2)	NOT NULL	Product price
vendor_id	INT	FOREIGN KEY → users(user_id)	Vendor who added the product
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Product creation timestamp

3. Orders

Column Name	Data Type	Data Type	Description
order_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each order
customer_id	INT	FOREIGN KEY → users(user_id)	Customer placing the order
product_id	INT	FOREIGN KEY → products(product_id)	Ordered product
vendor_id	INT	FOREIGN KEY → users(user_id)	Vendor fulfilling the order
status	ENUM	('placed','cancelled','shipped','delivered','returned') DEFAULT 'placed'	Current order status
order_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Order placement date

Relationships

- Users → Products: One vendor can add many products (**One-to-Many**).
- Users → Orders: One customer can place many orders (**One-to-Many**).
- Products → Orders: One product can appear in many orders (**One-to-Many**).

1.Users Table

```
CREATE TABLE users (  
  user_id INT PRIMARY KEY AUTO_INCREMENT,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  role ENUM('customer','vendor','admin') NOT NULL,  
  phone VARCHAR(10) UNIQUE NOT NULL,  
  address VARCHAR(255) NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
```

2. Products Table

```
CREATE TABLE products (  
    product_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100) NOT NULL,  
    description TEXT NOT NULL,  
    price DECIMAL(10,2) NOT NULL,  
    vendor_id INT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (vendor_id) REFERENCES users(user_id)  
);
```

3. Orders Table

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_id INT,  
    product_id INT,  
    vendor_id INT,  
    status ENUM('placed','cancelled','shipped','delivered','returned') DEFAULT 'placed',  
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (customer_id) REFERENCES users(user_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id),  
    FOREIGN KEY (vendor_id) REFERENCES users(user_id)  
);
```